Mathematical Methods for Finance

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Module 3 - Lesson 9

Probability, events and random variables

From uncertain outcomes to numerical quantities

Learning outcomes for Lesson 9

By the end of the lesson students should be able to

1. build a finite probability space and identify events;
2. use event operations, inclusion-exclusion and complements;
3. compute conditional probabilities and apply Bayes formula;
4. distinguish events, random variables, distributions and numbers;
5. compute expectation, variance and simple probabilities for standard models.

Intuition: Operating rule

Every formula must be attached to an object: event, random variable, distribution, expectation or variance.

How today is organised

Block 1

14:30–15:20
Probability spaces, events, operations and conditional probability.

Block 2

15:35–16:25
Random variables, distributions, expectation, variance and models.

Block 3

16:40–17:30
Guided exercises, independent exercise and exam-style checklist.

Exercise: During the lesson

For each exercise, first write the mathematical object involved before computing.

Block 1

Probability spaces and events

From measure to probability

Definition: Probability space

A probability space is a triple

$$(\Omega, \mathcal{F}, \mathbb{P}),$$

where Ω is the sample space, $\mathcal{F}$ is the family of events and $\mathbb{P}$ assigns probabilities.

Intuition: Translation

- outcome = one possible result;
- event = set of outcomes;
- probability = size of an event;
- random variable = measurable numerical summary.

A complete finite model

Example: Two coin tosses

$$\Omega = \{HH, HT, TH, TT\}, \qquad \mathcal{F} = 2^{\Omega}.$$

If the coin is fair, every outcome has probability $1/4$.

Exercise: Quick computation $\star$

Let X be the number of heads. Compute

$$\mathbb{P}(X = 0), \qquad \mathbb{P}(X = 1), \qquad \mathbb{P}(X \geq 1).$$

Worked example: Solution

$X = 0$ only for TT, so $\mathbb{P}(X = 0) = 1/4$. $X = 1$ for HT, TH, so $\mathbb{P}(X = 1) = 1/2$. Finally $\mathbb{P}(X \geq 1) = 3/4$.

Events and operations

Definition: Operations

For events $A, B \in \mathcal{F}$:

$$A \cap B, \qquad A \cup B, \qquad A^c, \qquad A \setminus B.$$

Example: Dice roll

$\Omega = \{1, 2, 3, 4, 5, 6\}$. Let

$$A = \{2, 4, 6\}, \quad B = \{4, 5, 6\}.$$

Exercise: Do it $\star$

Compute $A \cap B$, $A \cup B$, A^c and $A \setminus B$.

Basic probability rules

Theorem: Rules

For events A, B:

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A),$$

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

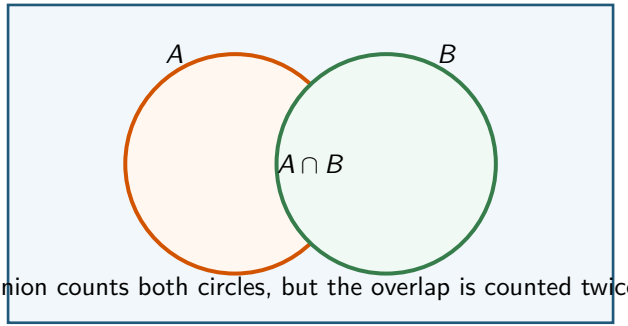
If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Counterexample: Most common mistake

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$$

is true only when $A \cap B = \emptyset$.

Visual: inclusion-exclusion



Union counts both circles, but the overlap is counted twice

Formula

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Worked exercise: missing intersection

Exercise: Exam-style ⋆⋆

Suppose

$$\mathbb{P}(A) = 0.55, \qquad \mathbb{P}(B) = 0.40, \qquad \mathbb{P}(A \cup B) = 0.70.$$

Compute $\mathbb{P}(A \cap B)$, $\mathbb{P}(A \setminus B)$, and decide whether A, B can be disjoint.

Worked example: Solution

By inclusion-exclusion,

$$\mathbb{P}(A \cap B) = 0.55 + 0.40 - 0.70 = 0.25.$$

Then

$$\mathbb{P}(A \setminus B) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = 0.30.$$

They cannot be disjoint because $\mathbb{P}(A \cap B) > 0$.

Conditional probability

Definition: Conditional probability

If $\mathbb{P}(B) > 0$, then

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Intuition: Meaning

Conditioning on B means that the sample space has been restricted to B. Probabilities are then renormalized inside B.

Exercise: Quick check $\star$

If $\mathbb{P}(A \cap B) = 0.18$ and $\mathbb{P}(B) = 0.60$, compute $\mathbb{P}(A \mid B)$.

Solution: conditional probability

Worked example: Direct computation

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \frac{0.18}{0.60} = 0.30.$$

Counterexample: Notation trap

$\mathbb{P}(A \mid B)$ is not symmetric in A and B. Usually

$$\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A).$$

Independence

Definition: Independent events

Two events A and B are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Equivalently, if $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \mathbb{P}(A).$$

Exercise: Exam-style $\star\star$

Let $\mathbb{P}(A) = 0.4$, $\mathbb{P}(B) = 0.5$, $\mathbb{P}(A \cap B) = 0.2$. Are A and B independent?

Worked example: Solution

$\mathbb{P}(A)\mathbb{P}(B) = 0.4 \cdot 0.5 = 0.2 = \mathbb{P}(A \cap B)$. Yes, they are independent.

Bayes formula

Theorem: Bayes formula

If $\mathbb{P}(B) > 0$, then

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

With a partition A, A^c:

$$\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A) + \mathbb{P}(B \mid A^c)\mathbb{P}(A^c).$$

Intuition: How to read it

Bayes formula reverses the conditioning direction: from $\mathbb{P}(B \mid A)$ to $\mathbb{P}(A \mid B)$.

Worked Bayes computation

Exercise: Exam-style ⋆⋆

Assume

$$\mathbb{P}(A) = 0.3, \qquad \mathbb{P}(B \mid A) = 0.8, \qquad \mathbb{P}(B \mid A^c) = 0.2.$$

Compute $\mathbb{P}(B)$ and $\mathbb{P}(A \mid B)$.

Worked example: Step-by-step solution

First,

$$\mathbb{P}(B) = 0.8 \cdot 0.3 + 0.2 \cdot 0.7 = 0.38.$$

Then

$$\mathbb{P}(A \mid B) = \frac{0.8 \cdot 0.3}{0.38} = \frac{0.24}{0.38} \approx 0.632.$$

Block 1 mini-workshop

Exercise: Three quick tasks

1. If $\mathbb{P}(A) = 0.7$ and $\mathbb{P}(B) = 0.6$, find the range of possible values for $\mathbb{P}(A \cap B)$.
2. Give an example of two events with $\mathbb{P}(A \mid B) \neq \mathbb{P}(B \mid A)$.
3. Explain in one sentence why disjoint events with positive probability are not independent.

Target skill

Recognize whether the problem is about complement, union, conditioning or independence.

Block 2

Random variables, distributions and moments

Random variables

Definition: Random variable

A random variable is a measurable map

$$X : \Omega \to \mathbb{R}.$$

It turns outcomes into numbers.

Example: Coin tosses

$X =$ number of heads.

Counterexample: Notation trap

$\mathbb{P}(X)$ is usually meaningless. Use events such as

$$\{X \leq x\}, \quad \{X = 2\}, \quad \{X > 0\}.$$

Object grammar

Classify expressions

Expression	Object
X	random variable
$X(\omega)$	number
$\{X > 3\}$	event
$\mathbb{P}(X > 3)$	number
$\mathbb{E}[X]$	number, if it exists

Exercise: Quick classification $\star$

Classify $X + Y$, $\{X \leq Y\}$, $\mathbb{E}[X^2 + Y^2]$, $\mathbb{P}(X \leq Y)$.

Distribution function

Definition: Cumulative distribution function

The distribution function of X is

$$F_X(x) = \mathbb{P}(X \leq x).$$

Theorem: Basic properties

- F_X is increasing;
- $\lim_{x \to -\infty} F_X(x) = 0$, $\lim_{x \to +\infty} F_X(x) = 1$;
- F_X is right-continuous.

Discrete random variables

Definition: Discrete law

A discrete random variable has probabilities

$$p_k = \mathbb{P}(X = x_k), \qquad \sum_k p_k = 1.$$

Exercise: Distribution table ⋆⋆

Let X take values $-1, 0, 2$ with probabilities $0.2, 0.5, 0.3$. Compute $\mathbb{P}(X > 0)$ and $\mathbb{P}(X \le 0)$.

Worked example: Solution

$\mathbb{P}(X > 0) = \mathbb{P}(X = 2) = 0.3$. Also $\mathbb{P}(X \le 0) = 0.2 + 0.5 = 0.7$.

Continuous random variables

Definition: Density

A continuous random variable has density f_X if

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x)\,dx.$$

Intuition: Area

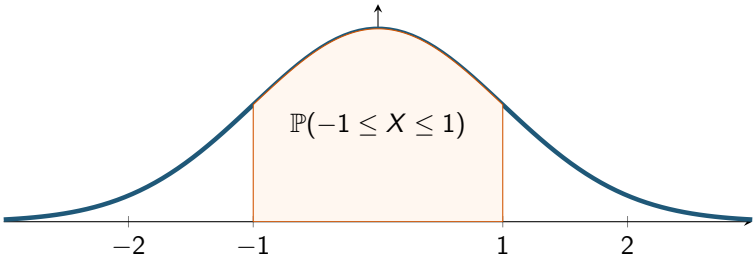
Probability is area under the density curve. For a continuous variable,

$$\mathbb{P}(X = a) = 0.$$

Counterexample: Common mistake

The value $f_X(a)$ is not a probability. It is a density height.

Visual: density and probability



Message

An interval probability is an area, not a height.

Expectation

Definition: Expected value

For a discrete random variable,

$$\mathbb{E}[X] = \sum_k x_k \mathbb{P}(X = x_k).$$

For a continuous random variable with density f_X,

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f_X(x)\, dx.$$

Intuition: Centre of mass

The expectation is a weighted average. It does not need to be a possible value of the random variable.

Worked expectation

Exercise: Discrete expectation ⋆⋆

Let X take values $-1, 0, 2$ with probabilities $0.2, 0.5, 0.3$. Compute $\mathbb{E}[X]$.

Worked example: Solution

$$\mathbb{E}[X] = (-1)0.2 + 0 \cdot 0.5 + 2 \cdot 0.3 = 0.4.$$

Counterexample: Interpretation

The value 0.4 is not one of the possible values of X. It is the long-run average predicted by the model.

Linearity of expectation

Theorem: Linearity

If expectations exist, then

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c.$$

No independence is required.

Exercise: Quick use $\star$

If $\mathbb{E}[X] = 3$ and $\mathbb{E}[Y] = -1$, compute $\mathbb{E}[2X - 4Y + 5]$.

Worked example: Solution

$$\mathbb{E}[2X - 4Y + 5] = 2 \cdot 3 - 4(-1) + 5 = 15.$$

Variance and standard deviation

Definition: Variance

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

The standard deviation is $\sqrt{\mathrm{Var}(X)}$.

Exercise: Compute variance $\star\star$

For the previous variable $X \in \{-1, 0, 2\}$, compute $\mathrm{Var}(X)$.

Worked example: Solution outline

$\mathbb{E}[X] = 0.4$. Also $\mathbb{E}[X^2] = 1 \cdot 0.2 + 0 + 4 \cdot 0.3 = 1.4$. Thus $\mathrm{Var}(X) = 1.4 - 0.16 = 1.24$.

Jensen inequality

Theorem: Jensen inequality

If φ is convex and the expectations exist, then

$$\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)].$$

Example: Square function

With $\varphi(x) = x^2$,

$$(\mathbb{E}[X])^2 \leq \mathbb{E}[X^2].$$

This is exactly $\mathrm{Var}(X) \geq 0$.

Important models: Bernoulli and binomial

Definition: Two basic models

- $X \sim \text{Bernoulli}(p)$: one trial, $X \in \{0, 1\}$.
- $Y \sim \text{Binomial}(n, p)$: number of successes in n independent trials.

Formulas

$$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1-p),$$

$$\mathbb{P}(Y = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Binomial exercise

Exercise: At least one success ⋆⋆

Let $Y \sim \text{Binomial}(5, 0.2)$. Compute $\mathbb{P}(Y \geq 1)$ using the complement.

Worked example: Solution

$$\mathbb{P}(Y \geq 1) = 1 - \mathbb{P}(Y = 0) = 1 - (0.8)^5.$$

Since $0.8^5 = 0.32768$,

$$\mathbb{P}(Y \geq 1) = 0.67232.$$

Important models: Poisson and normal

Definition: Poisson model

If $N \sim \text{Poisson}(\lambda)$, then

$$\mathbb{P}(N = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad \mathbb{E}[N] = \text{Var}(N) = \lambda.$$

Definition: Normal model

If $X \sim N(\mu, \sigma^2)$, then $\mathbb{E}[X] = \mu$, $\text{Var}(X) = \sigma^2$, and

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

Normal standardization exercise

Exercise: Standardize ⋆⋆

Let $X \sim N(10, 4)$. Rewrite $\mathbb{P}(X \leq 12)$ in terms of a standard normal variable Z.

Worked example: Solution

Here $\mu = 10$ and $\sigma = 2$. Therefore

$$\mathbb{P}(X \leq 12) = \mathbb{P}\left(\frac{X - 10}{2} \leq \frac{12 - 10}{2}\right) = \mathbb{P}(Z \leq 1).$$

One table to choose a model

Situation	Typical variable	Model
One yes/no outcome	0 or 1	Bernoulli
Number of successes in fixed trials	$0, \dots, n$	Binomial
Number of rare events in a period	$0, 1, 2, \dots$	Poisson
Continuous variable around a centre	real-valued	Normal

Exercise: Model recognition $\star$

For each situation above, write the support of the random variable.

Block 3

Guided and independent exercises

Guided exercise 1: events and conditioning

Exercise: Problem $\star\star$

A class contains 40 students. 24 attend the first review session, 18 attend the second, and 10 attend both.

1. What is the probability that a randomly selected student attended at least one session?
2. What is the probability that the student attended the first session given that they attended the second?
3. Are the two events independent?

Solution: guided exercise 1

Worked example: Step-by-step

Let A be first session and B second session. Then

$$\mathbb{P}(A) = \frac{24}{40}, \quad \mathbb{P}(B) = \frac{18}{40}, \quad \mathbb{P}(A \cap B) = \frac{10}{40}.$$

$$\mathbb{P}(A \cup B) = \frac{24 + 18 - 10}{40} = \frac{32}{40} = 0.8.$$

$$\mathbb{P}(A \mid B) = \frac{10/40}{18/40} = \frac{10}{18} = \frac{5}{9}.$$

Independence would require $10/40 = (24/40)(18/40) = 0.27$, false.

Guided exercise 2: distribution and moments

Exercise: Problem $\star\star$

A random variable X has distribution

x	0	1	3
$\mathbb{P}(X = x)$	0.2	0.5	0.3

Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, $\mathrm{Var}(X)$ and $\mathbb{P}(X \geq 1)$.

Solution: guided exercise 2

Worked example: Detailed calculation

$$\mathbb{E}[X] = 0 \cdot 0.2 + 1 \cdot 0.5 + 3 \cdot 0.3 = 1.4.$$

$$\mathbb{E}[X^2] = 0^2 \cdot 0.2 + 1^2 \cdot 0.5 + 3^2 \cdot 0.3 = 3.2.$$

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 3.2 - 1.96 = 1.24.$$

$$\mathbb{P}(X \geq 1) = 0.5 + 0.3 = 0.8.$$

Guided exercise 3: Bayes table

Exercise: Problem $\star\star\star$

A group is split into A and A^c with $\mathbb{P}(A) = 0.4$. A test is positive with probability 0.9 in group A and with probability 0.3 in group A^c. Compute the probability that a positive observation comes from group A.

Hint

First compute total probability of a positive result.

Solution: guided exercise 3

Worked example: Bayes formula

Let B be the event positive. Then

$$\mathbb{P}(B) = 0.9 \cdot 0.4 + 0.3 \cdot 0.6 = 0.36 + 0.18 = 0.54.$$

Therefore

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)} = \frac{0.36}{0.54} = \frac{2}{3}.$$

Independent exercise

Exercise: Try alone $\star\star\star$

A random variable Y takes values $-2, 1, 4$ with probabilities $0.25, 0.50, 0.25$.

1. Compute $\mathbb{E}[Y]$, $\mathrm{Var}(Y)$.
2. Compute $\mathbb{P}(Y > 0)$.
3. Let $Z = 2Y + 1$. Compute $\mathbb{E}[Z]$ without finding the whole distribution of Z.

Solution: independent exercise

Worked example: Solution

$$\mathbb{E}[Y] = -2(0.25) + 1(0.50) + 4(0.25) = 1.$$

$$\mathbb{E}[Y^2] = 4(0.25) + 1(0.50) + 16(0.25) = 5.5.$$

$$\mathrm{Var}(Y) = 5.5 - 1^2 = 4.5.$$

$$\mathbb{P}(Y > 0) = 0.50 + 0.25 = 0.75.$$

By linearity,

$$\mathbb{E}[Z] = \mathbb{E}[2Y + 1] = 2\mathbb{E}[Y] + 1 = 3.$$

Exit ticket

Exercise: Three questions before leaving

1. What is the difference between an event and a random variable?
2. What is the fastest way to compute $\mathbb{P}(\text{at least one})$?
3. Which formula tells whether two events are independent?

Answer in one sentence each

These questions reveal whether the lesson objects are clear before moving to random vectors.

End-of-lesson checklist

You should now be able to

- build a finite sample space and events;
- compute union, complement and conditional probabilities;
- apply Bayes formula without confusing conditioning directions;
- distinguish X, $X(\omega)$, $\{X \leq x\}$, $F_X(x)$;
- compute expectation and variance for a discrete law;
- recognize Bernoulli, binomial, Poisson and normal models.

Suggested practice after Lesson 9

Core exercises

From the Module 3 exercise pack: probability spaces, event operations, conditional probability, Bayes formula and basic discrete variables.

Exercise: Recommended self-training

Select:

- two exercises on event operations;
- two exercises on conditional probability;
- two exercises on random variable distributions;
- two exercises on expectation and variance.